

Grade 11 – Term 1

C4 Practice Activity 4 with Solutions

Contents

1	Introduction	1
2	Rate and Waiting-Time Interpretation	2
3	Waiting-Time Probabilities	3
4	Conditional Waiting Times and Memorylessness	5
5	Assessing Model Suitability	7
6	Applied Economic and Financial Problems	8

Evaluated criterion

- C4: Applies the exponential distribution to constant-rate waiting-time situations and assesses its suitability.

1 Introduction

The exponential distribution models a *non-negative waiting time* T until the next event when events occur independently at a reasonably stable average rate. If the average number of events per unit of time is $\lambda > 0$, then

$$T \sim \text{Exp}(\lambda), \quad T \geq 0,$$

and the pdf, CDF, and survival function are

$$f_T(t) = \begin{cases} \lambda e^{-\lambda t}, & t \geq 0, \\ 0, & t < 0, \end{cases} \quad F_T(t) = P(T \leq t) = 1 - e^{-\lambda t}, \quad P(T > t) = e^{-\lambda t}.$$

Consequently,

$$P(a < T < b) = F_T(b) - F_T(a) = e^{-\lambda a} - e^{-\lambda b}, \quad E(T) = \frac{1}{\lambda}.$$

Because T is continuous, a single endpoint has probability 0; for example, $P(T < 5) = P(T \leq 5)$. The units must match before a formula is used. Thus, a rate of 12 events per hour is $12/60 = 0.2$ event per minute.

The distribution is also *memoryless*:

$$P(T > s + t \mid T > s) = P(T > t).$$

Under the model, waiting s time units without an event does not make the event more nearly “due.” This feature is credible only while the mechanism and rate remain unchanged.

For each context, first define the waiting-time variable and its support. Then identify or determine λ , convert its units, and write the model. Select the CDF, survival, interval, or memoryless formula; show the substitution and calculation; and interpret the answer in context. Finally, check suitability: the event rate should be reasonably stable, events should be approximately independent, and the variable must genuinely be a time to the next event. Fixed schedules, changing demand, queue-dependent service, event clustering, and aging equipment can all make an exponential model unsuitable.

2 Rate and Waiting-Time Interpretation

Problems:

2.1. Fintech verification enquiries

2.1 Fintech verification enquiries

A Bogotá fintech records 144 account-verification enquiries during the 12 hours from 7:00 a.m. to 7:00 p.m. For a preliminary model, an analyst lets W be the number of minutes from a randomly chosen observation time until the next enquiry. The analyst proposes that enquiries occur independently at a constant average rate throughout the period.

Questions/Tasks.

C4.1 Identify the variable, its support, and the rate in enquiries per hour and per minute.

C4.2 Write the exponential model, including its pdf and CDF.

C4.3 Find and interpret the expected waiting time.

C4.4 Assess whether using the whole-period average is reasonable.

C4.1 - Identify the variable, support, and rate

The event is the arrival of the next verification enquiry. Therefore W is a waiting time measured in minutes, and its support is

$$w \geq 0.$$

First calculate the hourly rate:

$$\lambda = \frac{144 \text{ enquiries}}{12 \text{ hours}} = 12 \text{ enquiries per hour.}$$

Because W is measured in minutes, convert hours to minutes:

$$\lambda = \frac{12}{60} = 0.2 \text{ enquiry per minute.}$$

This is an average rate, not a claim that enquiries occur exactly five minutes apart.



C4.2 - Write the model, pdf, and CDF

Using the per-minute rate found in C4.1,

$$W \sim \text{Exp}(0.2), \quad W \text{ measured in minutes.}$$

Substitute $\lambda = 0.2$ into the general formulas. The pdf is

$$f_W(w) = \begin{cases} 0.2e^{-0.2w}, & w \geq 0, \\ 0, & w < 0, \end{cases}$$

and the CDF is

$$F_W(w) = P(W \leq w) = \begin{cases} 1 - e^{-0.2w}, & w \geq 0, \\ 0, & w < 0. \end{cases}$$

The support excludes negative waits, and the CDF gives the probability that the next enquiry has appeared by time w .



C4.3 - Find and interpret the mean wait

For an exponential waiting time,

$$E(W) = \frac{1}{\lambda}.$$

Substituting the per-minute rate gives

$$E(W) = \frac{1}{0.2} = 5 \text{ minutes.}$$

Across many comparable observation times, the mean wait to the next enquiry would be 5 minutes. Individual waits need not equal 5 minutes.



C4.4 - Assess the constant-rate assumption

The calculation distributes 144 enquiries across the entire 12-hour period. That model is reasonable only if shorter time blocks have similar rates and one customer's enquiry does not trigger or suppress another. Demand may instead rise at lunch, after work, or following a salary-payment message. In that case the whole-period average hides a changing rate, so one exponential model is unsuitable. The analyst should inspect counts by hour and use the model only within blocks whose rates are reasonably stable.



3 Waiting-Time Probabilities

Problems:

3.1. Off-peak courier dispatches

3.1 Off-peak courier dispatches

During a specified off-peak hour, a bicycle courier in Chapinero receives independent dispatch requests at a stable average rate of 4 per hour. Let T be the waiting time in minutes from the end of one job until the next request. Operations staff want to understand both short waits and costly idle periods.

Questions/Tasks.

C4.1 Write the model and find the probability that a request arrives within 10 minutes.

C4.2 Use a complement to find the probability that the wait exceeds 20 minutes.

C4.3 Find the probability that the wait is between 10 and 20 minutes.

C4.4 Interpret the three probabilities for the courier operation.

C4.1 - Model a wait of at most 10 minutes

Convert the hourly rate to the units of T :

$$\lambda = \frac{4}{60} = \frac{1}{15} \text{ request per minute,} \quad T \sim \text{Exp}\left(\frac{1}{15}\right).$$

“Within 10 minutes” means $T \leq 10$, so use the CDF:

$$\begin{aligned} P(T \leq 10) &= F_T(10) \\ &= 1 - e^{-(1/15)(10)} \\ &= 1 - e^{-2/3} \\ &\approx 1 - 0.5134 = 0.4866. \end{aligned}$$

Thus there is about a 48.7% chance that the next request arrives within 10 minutes.



C4.2 - Use a complement for a long wait

The event $T > 20$ is the complement of $T \leq 20$. Therefore

$$\begin{aligned} P(T > 20) &= 1 - P(T \leq 20) \\ &= 1 - \left(1 - e^{-(1/15)(20)}\right) \\ &= e^{-4/3} \\ &\approx 0.2636. \end{aligned}$$

Under the model, about 26.4% of comparable off-peak waits last more than 20 minutes.



C4.3 - Find an interval probability

For a continuous variable, including or excluding the endpoints does not change the probability. Subtract the two CDF values:

$$\begin{aligned}P(10 < T < 20) &= F_T(20) - F_T(10) \\&= \left(1 - e^{-4/3}\right) - \left(1 - e^{-2/3}\right) \\&= e^{-2/3} - e^{-4/3} \\&\approx 0.5134 - 0.2636 = 0.2498.\end{aligned}$$

Hence about 25.0% of waits are between 10 and 20 minutes.



C4.4 - Interpret the probability breakdown

The three disjoint ranges cover all possible waits:

$$P(T \leq 10) + P(10 < T \leq 20) + P(T > 20) \approx 0.4866 + 0.2498 + 0.2636 = 1.0000.$$

Thus approximately 48.7% of requests arrive in the first 10 minutes, 25.0% in the next 10 minutes, and 26.4% after 20 minutes. The last group quantifies potential idle time, but the figures apply only to the stated off-peak regime; they should not be transferred to a lunch rush without a new rate.



4 Conditional Waiting Times and Memorylessness

Problems:

4.1. Tourism website enquiries

4.1 Tourism website enquiries

A Bogotá hotel receives independent website enquiries at a stable rate of 3 per hour during a quiet late-evening window. At 10:00 p.m. the employee starts monitoring the inbox. By 10:20 p.m. no enquiry has arrived. Let T be the total wait in hours from 10:00 p.m. to the next enquiry.

Questions/Tasks.

C4.1 Use conditional probability to find the chance that the total wait exceeds 50 minutes.

C4.2 Use memorylessness to obtain the same result.

C4.3 Find the probability that an enquiry arrives in the next 15 minutes.

C4.4 Interpret memorylessness and state when it would be implausible.

C4.1 - Calculate from conditional probability

The observed 20 minutes are $20/60 = 1/3$ hour, and 50 minutes are $50/60 = 5/6$ hour. Since $\{T > 5/6\} \subset \{T > 1/3\}$,

$$\begin{aligned} P\left(T > \frac{5}{6} \mid T > \frac{1}{3}\right) &= \frac{P(T > 5/6)}{P(T > 1/3)} \\ &= \frac{e^{-3(5/6)}}{e^{-3(1/3)}} \\ &= e^{-5/2+1} = e^{-3/2} \\ &\approx 0.2231. \end{aligned}$$

Given the first 20 alert-free minutes, the modeled probability that the total wait passes 50 minutes is about 22.3%.



C4.2 - Apply memorylessness

To pass 50 total minutes after already waiting 20 minutes requires more than

$$50 - 20 = 30 \text{ additional minutes} = 0.5 \text{ hour.}$$

The memoryless property therefore gives

$$\begin{aligned} P(T > 50 \text{ min} \mid T > 20 \text{ min}) &= P(T > 30 \text{ additional min}) \\ &= e^{-3(0.5)} = e^{-1.5} \approx 0.2231. \end{aligned}$$

This agrees with the conditional-probability calculation in C4.1.



C4.3 - Find a next-quarter-hour probability

Fifteen minutes are $15/60 = 0.25$ hour. By memorylessness, the additional wait has the same $\text{Exp}(3)$ distribution. Use the complement of no enquiry in the next quarter-hour:

$$\begin{aligned} P(\text{enquiry in next 15 min} \mid T > 20 \text{ min}) &= 1 - P(\text{no enquiry in next 15 min}) \\ &= 1 - e^{-3(0.25)} \\ &= 1 - e^{-0.75} \\ &\approx 1 - 0.4724 = 0.5276. \end{aligned}$$

There is about a 52.8% modeled chance of an enquiry between 10:20 p.m. and 10:35 p.m.



C4.4 - Interpret and qualify memorylessness

The 20 elapsed minutes change the total clock time but not the modeled distribution of the remaining wait. In particular, the lack of an enquiry does not make one “due.” This interpretation is reasonable only if the late-evening rate remains stable and enquiries are approximately independent. It would be implausible near a scheduled advertising launch, at closing time, or when a tour group generates several related enquiries, because time or a previous event would then change the arrival mechanism.



5 Assessing Model Suitability

Problems:

5.1. Three operational waiting times

5.1 Three operational waiting times

A Colombian operations manager considers using an exponential model for each of the following variables:

- (i) the wait for a bus advertised to depart every 10 minutes;
- (ii) the time until the next overnight payment confirmation, when hourly counts fluctuate around the same level and customers act independently;
- (iii) the remaining lifetime of warehouse scanners whose failure risk rises as components wear out.

Questions/Tasks.

C4.1 Identify the waiting-time variable and relevant exponential assumption in each case.

C4.2 Assess the bus and payment-confirmation models.

C4.3 Assess the scanner model and relate aging to memorylessness.

C4.4 State what evidence the manager should examine before using a model.

C4.1 - Match variables to assumptions

All three variables are non-negative times: time to the next bus, time to the next confirmation, and time to scanner failure. However, that fact alone is not enough. The bus requires random arrivals at a stable rate; confirmations require approximately independent customer actions and a stable rate; and a scanner requires an approximately constant failure hazard, meaning its chance of failing in the next short interval does not depend on its present age.



C4.2 - Assess buses and confirmations

The bus model is unsuitable under the stated schedule. Departures are concentrated at fixed 10-minute marks rather than occurring randomly at a constant rate. A rider's remaining wait also depends on position within the timetable.

The payment-confirmation model is reasonable as an approximation for the restricted overnight period: the variable is a time to the next event, the description supports a stable rate, and separate customers support independence. The conclusion should not automatically be extended to busy daytime periods or batch-processed confirmations.



C4.3 - Assess aging scanners

An exponential lifetime assumes that a scanner which has survived for s hours has the same remaining-life distribution as a new scanner:

$$P(T > s + t \mid T > s) = P(T > t).$$

The stated wear makes failure risk increase with age. Survival to age s therefore changes the remaining-life distribution, contradicting memorylessness and a constant hazard. The exponential model is unsuitable for these aging scanners,

although it could be a useful approximation during an early period in which the hazard is observed to be nearly constant.



C4.4 - Identify evidence for a decision

The manager should compare event counts across equal time blocks, plot or summarize observed waiting times, and look for trends, clusters, and scheduled peaks. For scanners, failure rates should also be compared by equipment age. Roughly stable block rates and no clear dependence support an exponential approximation; systematic timetable peaks, customer batches, or increasing age-specific failure rates contradict it. Suitability is thus an empirical and contextual judgment, not a consequence of the variable merely being positive.



6 Applied Economic and Financial Problems

Problems:

6.1. Flower-export logistics decision

6.2. Digital-wallet alert threshold

6.1 Flower-export logistics decision

A Bogotá flower exporter studies urgent pickup requests during a stable off-peak window. Historical records show a mean waiting time of 24 minutes between requests. The exporter may lose a sale if the next request does not appear within 30 minutes, and a dispatcher wants a probability estimate for staffing decisions. Let X be the wait in hours to the next request.

Questions/Tasks.

C4.1 Determine λ from the mean and write the model.

C4.2 Find the probability that the wait exceeds 30 minutes.

C4.3 Find the probability that the request arrives between 12 and 30 minutes.

C4.4 Interpret the results and assess their use during Valentine's week.

C4.1 - Determine the rate from the mean

Convert the mean to the hours used for X :

$$24 \text{ minutes} = \frac{24}{60} = 0.4 \text{ hour.}$$

Since $E(X) = 1/\lambda$,

$$\frac{1}{\lambda} = 0.4 \quad \implies \quad \lambda = \frac{1}{0.4} = 2.5 \text{ requests per hour.}$$

Therefore the proposed model is

$$X \sim \text{Exp}(2.5), \quad x \geq 0,$$

with X measured in hours.



C4.2 - Calculate the risk of a wait over 30 minutes

Thirty minutes are $30/60 = 0.5$ hour. Using the survival function,

$$\begin{aligned}P(X > 0.5) &= e^{-\lambda(0.5)} \\&= e^{-2.5(0.5)} \\&= e^{-1.25} \\&\approx 0.2865.\end{aligned}$$

Thus about 28.7% of modeled off-peak waits exceed 30 minutes.



C4.3 - Calculate an operational interval

Convert the lower endpoint as well:

$$12 \text{ minutes} = \frac{12}{60} = 0.2 \text{ hour}.$$

Then

$$\begin{aligned}P(0.2 < X < 0.5) &= e^{-2.5(0.2)} - e^{-2.5(0.5)} \\&= e^{-0.5} - e^{-1.25} \\&\approx 0.6065 - 0.2865 = 0.3200.\end{aligned}$$

The model assigns about a 32.0% chance to a next request arriving after 12 minutes but before 30 minutes.



C4.4 - Interpret and qualify the staffing evidence

The 28.7% long-wait probability quantifies the chance associated with the 30-minute business threshold, while the 32.0% interval probability describes a substantial middle group of waits. Neither probability alone determines staffing: the exporter must also compare labour cost, response rules, and the value of a lost sale.

The off-peak model should not be transferred unchanged to Valentine's week. Deadlines and promotions can raise the rate at different times and make requests arrive in clusters, violating stability and independence. Separate rates for reasonably stable time blocks would be more defensible.



6.2 Digital-wallet alert threshold

During a stable overnight window, a Colombian digital wallet models the wait Y to the next suspicious-transaction alert as exponential. Risk managers report that the probability of no alert during a 30-minute period is 0.40. Time Y will be measured in hours. An analyst has already monitored for 45 minutes without an alert.

Questions/Tasks.

C4.1 Use the reported probability to determine λ and the mean wait.

C4.2 Find the probability of an alert during the next hour, given the observed 45 alert-free minutes.

C4.3 Find the probability that the total wait is between 1 and 2 hours, given the observation.

C4.4 Interpret the results and assess the model during a coordinated fraud wave.

C4.1 - Determine the parameter from a probability

Thirty minutes are 0.5 hour. “No alert” means that the waiting time exceeds 0.5 hour, so

$$P(Y > 0.5) = e^{-0.5\lambda} = 0.40.$$

Take natural logarithms and solve:

$$-0.5\lambda = \ln(0.40) \quad \Rightarrow \quad \lambda = \frac{-\ln(0.40)}{0.5} \approx 1.8326 \text{ alerts per hour.}$$

Hence

$$Y \sim \text{Exp}(1.8326), \quad E(Y) = \frac{1}{\lambda} = \frac{0.5}{-\ln(0.40)} \approx 0.5457 \text{ h.}$$

Converting the mean,

$$0.5457(60) \approx 32.7 \text{ minutes.}$$

Thus the reported half-hour survival probability corresponds to a mean wait of about 32.7 minutes.



C4.2 - Find the next-hour conditional probability

The elapsed 45 minutes are 0.75 hour. By memorylessness, the probability of an alert in the next hour is the same as it was at the beginning:

$$\begin{aligned} P(Y \leq 1.75 \mid Y > 0.75) &= P(\text{additional wait} \leq 1) \\ &= 1 - e^{-\lambda(1)} \\ &= 1 - e^{-1.8326} \\ &\approx 1 - 0.1600 = 0.8400. \end{aligned}$$

There is about an 84.0% modeled chance of an alert in the coming hour.



C4.3 - Find a conditional interval probability

Given survival to 0.75 hour, a total wait from 1 to 2 hours corresponds to an additional wait from

$$1 - 0.75 = 0.25 \text{ hour} \quad \text{to} \quad 2 - 0.75 = 1.25 \text{ hours.}$$

Memorylessness and the interval formula give

$$\begin{aligned} P(1 < Y < 2 \mid Y > 0.75) &= e^{-\lambda(0.25)} - e^{-\lambda(1.25)} \\ &= e^{-1.8326(0.25)} - e^{-1.8326(1.25)} \\ &\approx 0.6325 - 0.1012 = 0.5313. \end{aligned}$$

Thus, conditional on the observed 45 alert-free minutes, about 53.1% of modeled cases place the next alert between one and two hours after monitoring began.



C4.4 - Interpret and assess the alert model

The next-hour probability can inform on-call coverage, while the conditional interval probability describes when within the overall monitoring session an alert may occur. These are probabilities under the model, not guarantees.

During a coordinated fraud wave, alerts may arrive in bursts and their rate may rise as the attack develops. Both independence and a stable rate would then fail, so the memoryless exponential model and the calculated probabilities would be unreliable. Managers should update the rate over shorter stable windows or use a model that allows clustering and a changing rate.

